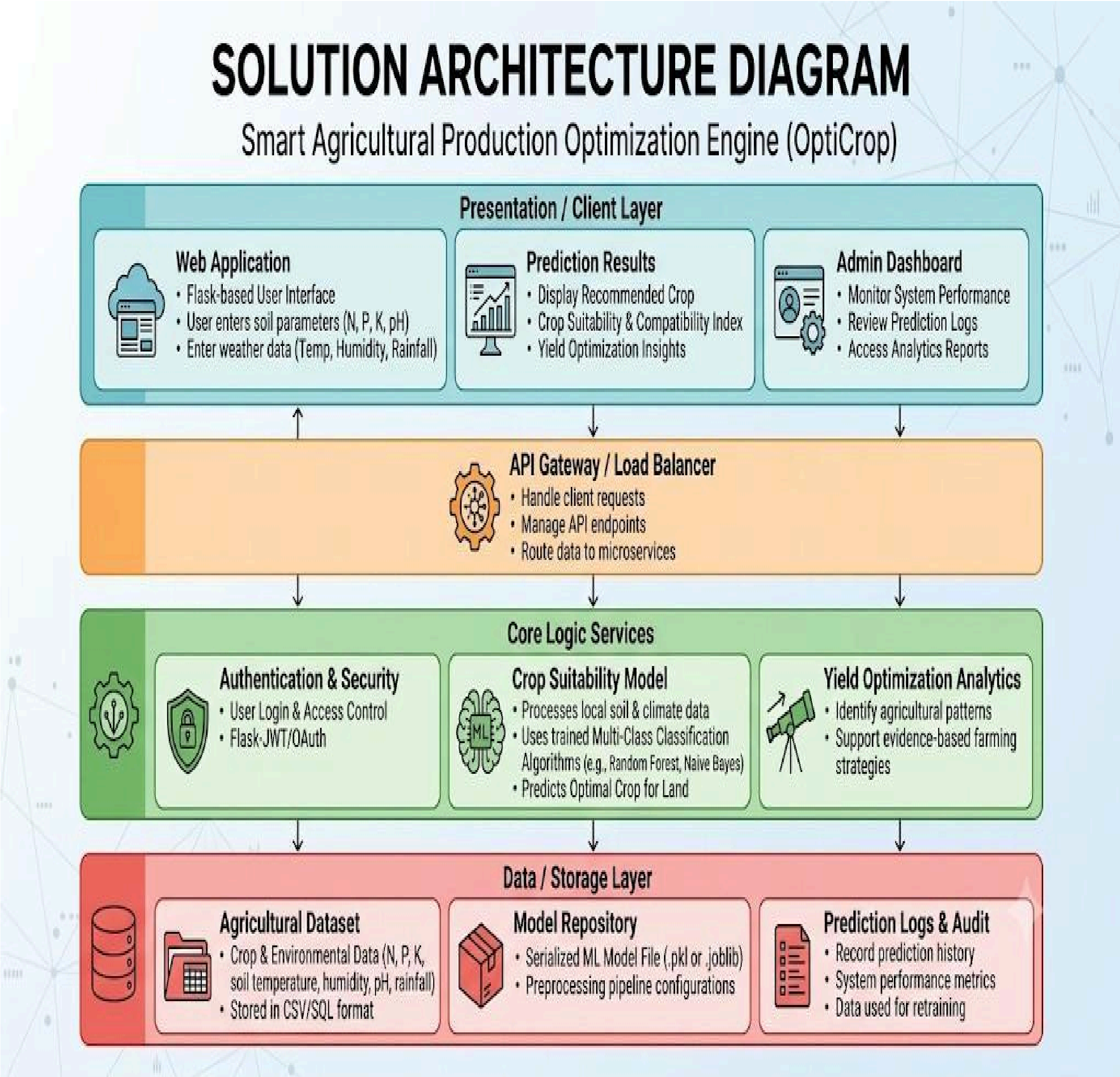
Solution Architecture

Date	28 June, 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description/Role	Technologies Used
Presentation Layer	Collects soil nutrient values (N, P, K), temperature, humidity, pH, and rainfall from the user and displays the recommended crop.	HTML, CSS,Bootstrap
Application Layer	Receives user input, validates the data, and communicates with the machine learning prediction model. .	Flask, Python
Machine Learning Layer	Analyzes soil and environmental parameters and predicts the most suitable crop.	Scikit-learn, Python
Data layer	Stores the crop dataset and trained machine learning model used for prediction.	CSV Dataset, Pickle(.pkl)